



VIETNAM NATIONAL UNIVERSITY HCMC

INTERNATIONAL UNIVERSITY

School of Computer Science and Engineering

Principle of Database Management

Final Report

Hotel Management System

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Table 1: Member's name and contribution

Course: Principle of Database Management

Academic advisor: Assoc. Prof. Nguyen Thi Thuy Loan

Project link: [Github](#)



FINAL REPORT

STUDENT RESEARCH PROJECT

Hotel Management System

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1. INTRODUCTION

1.1. Abstraction:

The tourism industry is booming, and to thrive in this competitive landscape, hotels need to innovate and streamline operations. The Hotel Management System (HMS) project equips you with 4 crucial steps to build a strong foundation for exceptional guest experiences and optimized hotel management.

- *Mapping the Landscape:* Begin by using entity-relationship diagrams (ERDs) to map key sample elements like rooms, guests, staff, and services. This establishes a foundation for efficient data storage.
- *Structuring the Database:* Organize data effectively through database normalization. This minimizes redundancy and ensures data integrity, leading to faster access and fewer errors.
- *Optimizing Queries:* Develop efficient SQL queries to retrieve data swiftly and accurately. This empowers staff to efficiently search reservations, guest details, and other critical information.
- *Integrating Data Sources:* Unify various data sources, like online bookings and inventory systems, to gain a comprehensive view of hotel operations. This allows for data-driven decision making.

The Hotel Management System (HMS) prioritizes both user experience and functionality. Guests can efficiently search for and reserve rooms, while staff can manage reservations, track availability, and streamline communication. Secure payment processing and intuitive interfaces for both staff and guests ensure a smooth experience.

This project equips students and professionals with practical experience in managing real-world hotel data, building scalable and efficient systems, and implementing best practices for data integrity, security, and performance

1.2. System Overview:

The hospitality industry is embracing the digital age, with Hotel Management Systems (HMS) becoming essential tools. Just like online marketplaces, HMS connects hotels with guests in a convenient.

Guests benefit from seamless booking, reservation management, and easy access to information, while hotels gain streamlined operations and improved communication.

This digital transformation empowers hotels to not only enhance guest experiences but also thrive in the competitive landscape. By embracing HMS, hotels can unlock a future of efficiency, guest satisfaction, and a competitive edge.

1.3. Goal:

- *Data Integrity:*
 - Ensure accurate and consistent guest information by preventing data errors and inconsistencies.

- Maintain a reliable single source of truth for all guest data.

➤ *Guest Communication:*

- Efficiently store and manage guest communication data, including inquiries and complaints.

- Facilitate smooth and timely interactions with guests through various channels.

➤ *Order Management:*

- Effectively manage reservations, track room availability, and streamline booking processes.

- Offer guests a seamless and efficient booking experience.

2. TASK TIMELINE & DIVISION

2.1. Contribution

Name	Responsibility
Trần Quang Phước	Database developer (main)
Đỗ Minh Hiếu	Database developer (main)
Nguyễn Mai Anh Nam	UI designer, Database developer(vice)
Lê Thị Hậu	UI designer, Database developer (vice)
Nguyễn Trần Bảo Khang	Database developer (main)
Nguyễn Nam	Database(vice), UI developer
Trần Thanh Phương	Database(vice), UI developer
Nguyễn Anh Minh	Database developer (main)

Table 2: Individual responsibility

2.2. Project Timeline & Task Division

Stage	Task	Member
PLANNING	Research information and experience about famous hotel management platform	Phước, Hiếu
	Collect sample data for analyzing	All
	Research for reference and technical documentation	All

	Unanimously decide on which communication, workflow, and tools to use	All
	Determine Scopes, objectives, and goals for project	All
	Analyze proper for approach and methodology for project development	All
	Specify requirements for the project	All
	Create timeline table and division task	Hiếu
	Determine proper technology stack and database management system	All
	Determine features and their priorities	All
	Proposal report	Hiếu, Anh Nam, Nam
	Phase 1 review and retrospective	All
CONCEPTUAL DESIGN	Identify use case and actors of the system	All
	Specify types of information that are essential	All
	Design user interface for core functionalities	Phước
	Design the relational models	All
	Design E.R diagram and class diagram	Hiếu. Phước
	Midterm report	All
	Phase 2 review and retrospective	All
IMPLEMENTING	Database creation and setup	All
	Develop web user interface and client-side functionality	Khang
	Work on new registration	Hậu

	Work on user login/logout and authentication	Phường
	Build dashboard website for user	Phước
	Develop login credential of application	Khang
	Develop dashboard implement for manager including showing list of booking, employees, customers, room/room type, and payment	Minh
	Develop employee management that showing the number of employees and roles of each employee	Hiếu
	Develop room management that showing list of rooms attach to type of rooms and price per room	Anh Nam
	Develop booking management that display all pending reservation from user and add the option to create/check for guest coming to hotel	Phước
	Develop payment management that check which booking have paid or not	Nguyễn Nam
	Phase 3 review and retrospective	All
TESTING	Code review and refactoring	All
	Bug detection and fixing	Phước
	Design and implement test case	Phước
	Phase 4 review and retrospective	All
PRESENTATION	Final report	Anh Nam, Phước, Hiếu
	Slides report	Hậu
	Listing requested information and finishing report	All

Table 3: Sprint planning and task division for individuals and teams

3. PROJECT ANALYSIS

Project link: [Github](#)

3.1. Requirement analysis

The Hotel Management System provides with the following functionalities:

- **Make a reservation:** Make a reservation for a customer by taking all the information about customers and by checking the availability of rooms in the hotel.
- **Room Status:** User can check the status of the room in the hotel ie. the number of rooms which are vacant, and which are non-vacant.
- **Check in and use of facilities:** After making reservation, customer can check-in in the hotel and can use all the facilities available in the hotel like restaurant, laundry, tea etc.
- **Check out:** Now the staff can calculate the total bill of customers and show it to customers at the time of customer checkout.

3.2. Approach analysis

Reviewed materials

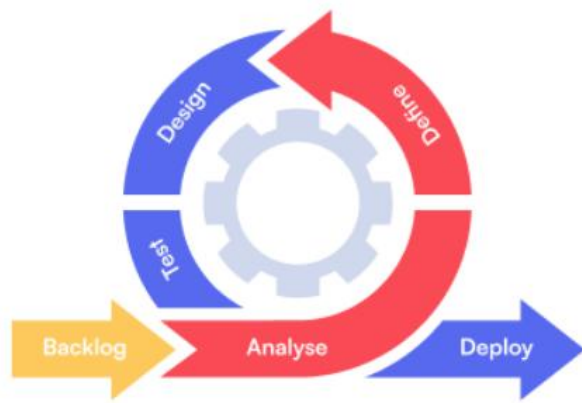
- + HTML/CSS/JavaScript/PHP/MySQL/Java (JRE 8/JDK 22.0 above)
- + IDE: Visual Studio/ NetBeans
- + Intermediate SQL Server language structure ("SQL Tutorial," n.d.)

Research Approach

● *Methodology*

Agile methodology is a way of managing projects that focuses on delivering quick results, being able to handle changing requirements and emphasizing teamwork. It encourages an iterative approach, with frequent reviews and adjustments, so that the end product meets the customer's needs.

● *Stages of the Agile software development lifecycle*



Agile methodology

Image 1: Agile methodology

3.3. System analysis Database Design

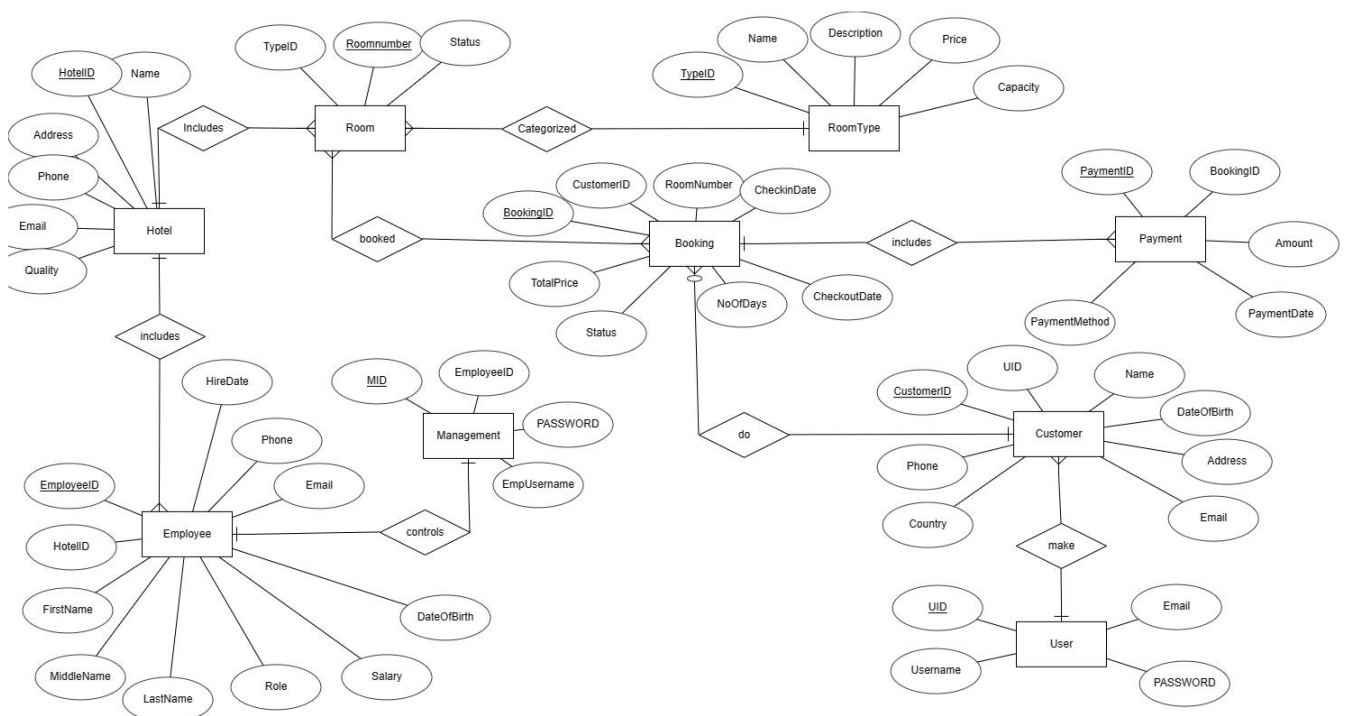


Image 2: Primary Crow's Foot Format E.R.D. Diagram

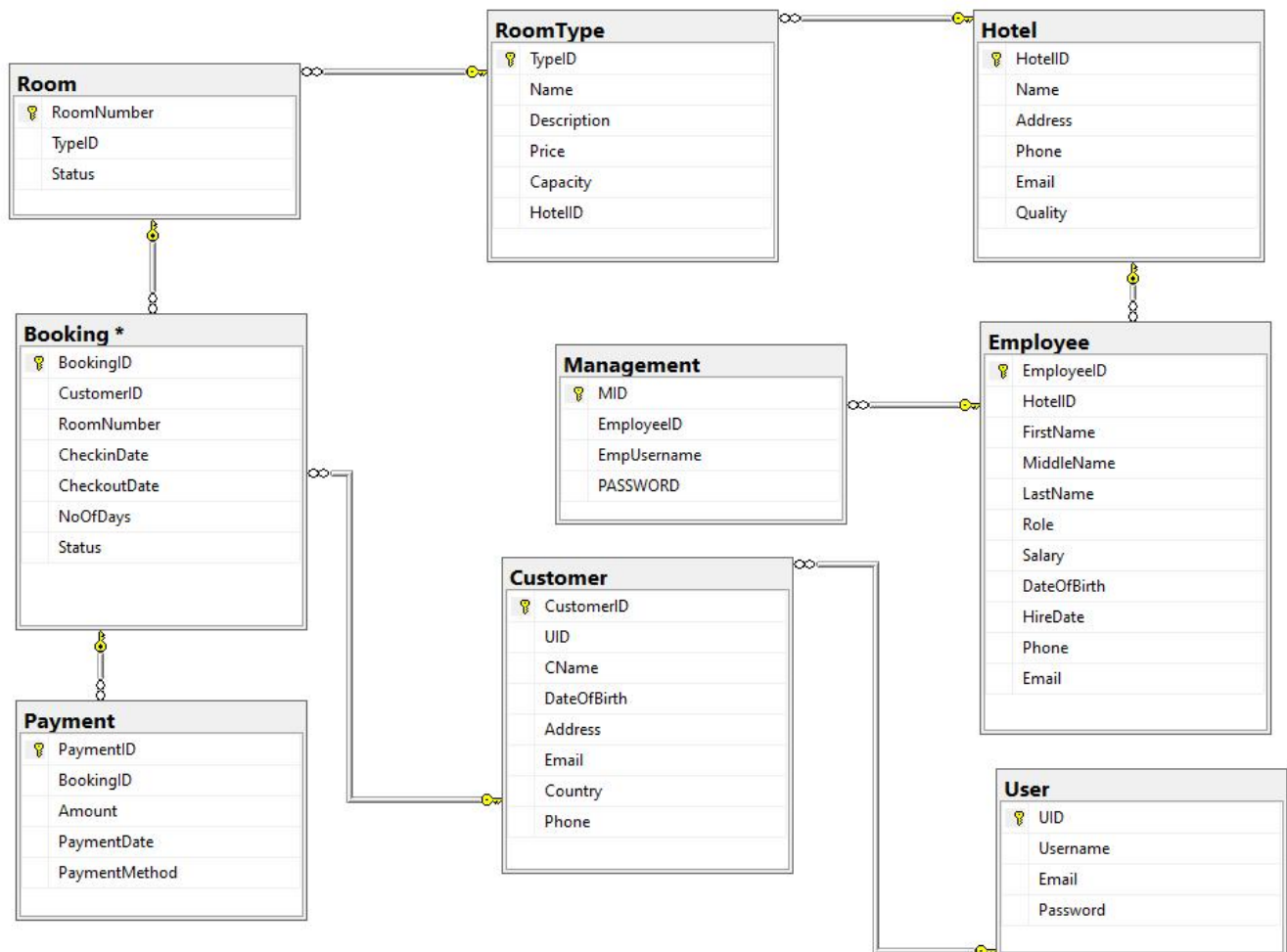


Image 3: Schema design generated by SQL Server Management Studio

Database Normalization

Main purpose

Data normalization is the process of adjusting the structure of a database to eliminate duplicate, unnecessary, or inconsistent data, while ensuring the consistency and integrity of the data within the database. Here how we apply data normalization into our Hotel Management Database System:

Normalization process

Step 1: 1st Normal Form (1NF)

- A single cell must not hold more than one value (atomicity).
- There must be a primary key for identification.
- No duplicated rows or columns.

Step 2: 2nd Normal Form (2NF): A table is in the second normal form if it is in the first normal form and every non-key attribute is fully functionally dependent on the primary key

Step 3: 3rd Normal Form (3NF): A table is in the third normal form if it is in the second normal form and there are no transitive functional dependencies.

Step 4: Boyce-Codd Normal Form (BCNF): Each determinant in a table should be a candidate key.

Conclusion: after following these steps above the database is 3NF.

3.4. Database and Table Creation

This part discusses the database creation, schemas, and tables following the E.R.D, the diagram in SQL Server Database:

Database Creation:

```
CREATE DATABASE HMS;  
  
USE HMS;
```

Tables Analysis:

The current version of the Entity-Relationship Diagram (E.R.D.) has the following entities:

Table 4: An overview of all database entities and their attributes

Table	Attributes	Explain
Hotel	HotelID INT PRIMARY KEY default 10, Name VARCHAR(50) default 'HCMIU', Address VARCHAR(100) default 'Quarter 6, Linh Trung ward, Thu Duc, HCMC', Phone VARCHAR(15) default '(028) 37244270', Email VARCHAR(100) default 'info@hcmiu.edu.vn', Quality INT default 5	The information of the hotel about name, address, phone, email, and hotel star rating(quality) for contact.
RoomType	TypeID INT PRIMARY KEY not null auto_increment, Name VARCHAR(50), Description VARCHAR(255),	Store type of room with details such as name, description, price, capacity, and hotel which belong to.

	Price DECIMAL(10, 2), Capacity INT, HotelID INT default 10, FOREIGN KEY (HotelID) references Hotel(HotelID)	
Room	RoomNumber INT PRIMARY KEY not null auto_increment, TypeID INT, Status VARCHAR(20), FOREIGN KEY (TypeID) REFERENCES RoomType(TypeID)	Showing what room is available or not and type of this room
Employee	EmployeeID INT not null PRIMARY KEY auto_increment, HotelID INT default 10, FirstName VARCHAR(25), MiddleName VARCHAR(25), LastName VARCHAR(25) NOT NULL, Role VARCHAR(25) NOT NULL, Salary DECIMAL(10, 2), DateOfBirth DATE, HireDate DATE, Phone VARCHAR(15), Email VARCHAR(255), FOREIGN KEY (HotelID) REFERENCES Hotel(HotelID)	Store information of individual employees who work in the hotel like their employee ID, their name, role, salary, date of birth, hire date, phone, email.
Management	MID INT PRIMARY KEY not null auto_increment, EmployeeID INT NOT NULL, EmpUsername VARCHAR(10), PASSWORD VARCHAR(64) NOT NULL, FOREIGN KEY (EmployeeID)	Provide manager credential to manage almost function like manage employee, booking, customer, payment in application.

	REFERENCES Employee(EmployeeID)	
User	UID INT PRIMARY KEY not null auto_increment, Username VARCHAR(16) NOT NULL, Email VARCHAR(32) NOT NULL, Password VARCHAR(32) NOT NULL	User credential who want to book room online.
Customer	CustomerID INT PRIMARY KEY not null auto_increment, UID INT NOT NULL, CName VARCHAR(50) NOT NULL, DateOfBirth DATE, Address VARCHAR(70), Email VARCHAR(70), Country VARCHAR(20), Phone VARCHAR(15), FOREIGN KEY (UID) REFERENCES User(UID)	Store information of guest that they want to register room reservation. This information includes the order number, name, birth, address, email, country, phone and account they are using for reservation.
Booking	BookingID INT PRIMARY KEY auto_increment, CustomerID INT NOT NULL, RoomNumber INT NOT NULL, CheckinDate DATE, CheckoutDate DATE, NoOfDays INT, Status VARCHAR(10), FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID), foreign key (RoomNumber) references	Record reservation details, including room number, check-in, check-out, and need receptionist review the booking status.

	Room(RoomNumber)	
Payment	PaymentID INT PRIMARY KEY auto_increment, BookingID INT NOT NULL, Amount DECIMAL(10, 2), PaymentDate DATE, PaymentMethod VARCHAR(50), FOREIGN KEY (BookingID) REFERENCES Booking(BookingID)	Handle payment information associated with reservation, including payment method, payment date, and payment value.

3.5. Database Data Insertion

Management table

```
INSERT INTO `Management` VALUES (1,1,'hcmiu','1234');
```

Employee table

```
INSERT INTO `Employee` VALUES (1,10,'PDM','Hotel','Group 6','manager',2000.00,'2000-11-11','2020-11-11','0882837487','hcmiu@gmail.com');
INSERT INTO `Employee` VALUES (2,10,'John','Lee','Receptionist',NULL,'2000-05-11','2024-05-01','0564392768','Lee@gmail.com');
INSERT INTO `Employee` VALUES (3,10,'Martin','Odegaard','Chef',NULL,'2000-05-10','2024-05-01','0536159753','o8@gmail.com');
INSERT INTO `Employee` VALUES (4,10,'Kai','Harvert','Waiter',NULL,'2000-05-13','2024-05-01','0264563289','King@gmail.com');
INSERT INTO `Employee` VALUES (5,10,'Bukayo','Saka','Housekeeping',NULL,'2000-05-17','2024-05-01','0456987465','spicy@gmail.com');
INSERT INTO `Employee` VALUES (6,10,'Declan','Rice','Reservation',NULL,'2000-05-03','2024-05-01','0365426987','100m@gmail.com');
INSERT INTO `Employee` VALUES (7,10,'William','Saliba','Bell man',NULL,'2000-05-03','2024-05-01','0365426987','Salibo@gmail.com');
```

Hotel table

```
INSERT INTO `Hotel` VALUES (10,'HCMIU','Quarter 6, Linh Trung ward, Thu Duc, HCMC','(028) 37244270','info@hcmiu.edu.vn',5);
```

RoomType table

```
INSERT INTO `RoomType` VALUES (1,'Superior Room','King room ',300.00,4,10);
INSERT INTO `RoomType` VALUES (2,'Superior Room','excellent',300.00,4,10);
INSERT INTO `RoomType` VALUES (3,'Queen Room','excellent',200.00,3,10);
INSERT INTO `RoomType` VALUES (4,'Queen Room','excellent',200.00,3,10);
INSERT INTO `RoomType` VALUES (5,'Single Room','good',30.00,1,10);
INSERT INTO `RoomType` VALUES (6,'Single Room','good',30.00,1,10);
INSERT INTO `RoomType` VALUES (7,'Single Room','good',30.00,1,10);
INSERT INTO `RoomType` VALUES (8,'Studio Room','good',60.00,3,10);
INSERT INTO `RoomType` VALUES (9,'Studio Room','nice',60.00,3,10);
INSERT INTO `RoomType` VALUES (10,'Deluxe Room','very nice',100.00,3,10);
INSERT INTO `RoomType` VALUES (11,'Queen Room','excellent',200.00,2,10);
INSERT INTO `RoomType` VALUES (12,'Deluxe Room','very nice',100.00,3,10);
INSERT INTO `RoomType` VALUES (13,'Deluxe Room','very nice',100.00,4,10);
INSERT INTO `RoomType` VALUES (14,'Single Room','normal',30.00,1,10);
INSERT INTO `RoomType` VALUES (15,'Single Room','normal',30.00,1,10);
INSERT INTO `RoomType` VALUES (16,'Studio Room','',60.00,2,10);
INSERT INTO `RoomType` VALUES (17,'Studio Room','good',60.00,2,10);
INSERT INTO `RoomType` VALUES (18,'Superior Room','King room',300.00,1,10);
INSERT INTO `RoomType` VALUES (19,'Deluxe Room','nice',100.00,2,10);
INSERT INTO `RoomType` VALUES (20,'Single Room','nice',30.00,2,10);
INSERT INTO `RoomType` VALUES (21,'Single Room','nice',30.00,2,10);
INSERT INTO `RoomType` VALUES (22,'Single Room','nice',30.00,2,10);
INSERT INTO `RoomType` VALUES (23,'Superior Room','good',300.00,1,10);
INSERT INTO `RoomType` VALUES (24,'Superior Room','good',300.00,1,10);
INSERT INTO `RoomType` VALUES (25,'Single Room','',30.00,1,10);
```

Room Table

```
INSERT INTO `Room` VALUES (2,2,'Ready');
INSERT INTO `Room` VALUES (3,3,'Ready');
INSERT INTO `Room` VALUES (4,4,'Ready');
INSERT INTO `Room` VALUES (5,5,'Ready');
INSERT INTO `Room` VALUES (6,6,'Ready');
INSERT INTO `Room` VALUES (7,7,'Ready');
INSERT INTO `Room` VALUES (8,8,'Ready');
INSERT INTO `Room` VALUES (9,9,'Ready');
INSERT INTO `Room` VALUES (10,10,'Ready');
INSERT INTO `Room` VALUES (11,11,'Ready');
INSERT INTO `Room` VALUES (12,12,'Ready');
INSERT INTO `Room` VALUES (13,13,'Ready');
INSERT INTO `Room` VALUES (14,14,'Ready');
INSERT INTO `Room` VALUES (15,15,'Used');
INSERT INTO `Room` VALUES (16,16,'Ready');
INSERT INTO `Room` VALUES (17,17,'Ready');
INSERT INTO `Room` VALUES (18,18,'Ready');
INSERT INTO `Room` VALUES (19,19,'Ready');
INSERT INTO `Room` VALUES (20,20,'Ready');
INSERT INTO `Room` VALUES (21,21,'Cleaning');
INSERT INTO `Room` VALUES (22,22,'Ready');
INSERT INTO `Room` VALUES (23,23,'Ready');
INSERT INTO `Room` VALUES (24,24,'Cleaning');
INSERT INTO `Room` VALUES (25,25,'Cleaning');
```

Customer Table

```
INSERT INTO `Customer` VALUES (4,4,'Gabriel Jesus','2000-05-16','Arsenal','js@gmail.com','Brazil','0564324643');
INSERT INTO `Customer` VALUES (5,4,'Thomas Partey','2000-05-21','Arsenal','TP@gmail.com','Ghana','0546982317');
INSERT INTO `Customer` VALUES (6,5,'Nguyen Nguyen','1997-02-27','152 Tran Phu, Nha Trang','nguyennguyentran@gmail.com','Vietnam','09873628918');
INSERT INTO `Customer` VALUES (7,4,'Gabriel Martinelli','2000-05-17','Arsenal','GM@gmail.com','Brazil','0569846231');
INSERT INTO `Customer` VALUES (19,1,'Viet Nam','2000-05-11','Thu Duc','abc@gmail.com','Vietnam','0192836475');
INSERT INTO `Customer` VALUES (23,4,'Zichenko','2000-05-15','Arsenal','z30@gmail.com','Afghanistan','0564978356');
INSERT INTO `Customer` VALUES (25,2,'William','1980-01-01','LA','william@gmail.com','United States','0123524168');
INSERT INTO `Customer` VALUES (26,4,'Gabriel','2000-05-01','Arsenal','G@gmail.com','Brazil','569846231');
INSERT INTO `Customer` VALUES (27,4,'Jorginho','2000-05-22','Arsenal','J20@gmail.com','Italy','0579862341');
INSERT INTO `Customer` VALUES (28,4,'Benjamin White','2000-05-16','Arsenal','b4@gmail.com','United Kingdom','0453261789');
INSERT INTO `Customer` VALUES (29,4,'David Raya','2000-05-10','Arsenal','gk@gmail.com','Spain','0154693741');
INSERT INTO `Customer` VALUES (30,2,'Alice','1989-10-
```

```

25','Pakis','a@gmail.com','Pakistan','3216410629');
INSERT INTO `Customer` VALUES (31,5,'Eric Krylova','1985-12-26','Vico Giganti, 13, 06049-Bazzano
Inferiore PG','EricKrylova@jourrapide.com','Russian Federation','+39 366 0158925');
INSERT INTO `Customer` VALUES (32,5,'Đào Kim Tâm','2002-05-26','13/65 Nguyen Thai Hoc, Q1,
HCMC','DaoKimTam@dayrep.com','Vietnam','0908497879');
INSERT INTO `Customer` VALUES (33,5,'Rahmana Viderman','1998-01-09','3535 Driftwood Road
Oakland, CA 94612','RahmanaViderman@jourrapide.com','United States','+1 408-403-7158');
INSERT INTO `Customer` VALUES (34,5,'Harald Andreuzzi','2004-11-29','2311 Jerry Toth Drive Kodiak,
AK 99615','HaraldAndreuzzi@dayrep.com','United States','+1 907-486-9747');
INSERT INTO `Customer` VALUES (35,5,'Julia Kaitainen','1987-11-02','51, place de Miremont 47300
VILLENUEVE-SUR-LOT','JuliaKaitainen@armyspy.com','France','+33576561911');
INSERT INTO `Customer` VALUES (36,5,'Mee Pan','1992-01-27','73, rue du Gue Jacquet 36000
CHÂTEAUROUX','MeePan@teleworm.us','France','+33218412541');
INSERT INTO `Customer` VALUES (37,8,'Rinkimirikakuta','1997-02-06','Belgrad rkp.
25.','Rinkimirikakuta@teleworm.us','Hungary','9434795');
INSERT INTO `Customer` VALUES (38,8,'Bowman Brownlock','1950-03-15','3876 rue Parc Sherbrooke,
QC J1H 1R6','BowmanBrownlock@dayrep.com','Canada','8195789160');
INSERT INTO `Customer` VALUES (39,9,'Johny Deep','1999-08-01','345 VoVanNgan
','nganhminh23@gmail.com','Vietnam','0556783241');
INSERT INTO `Customer` VALUES (40,9,'Pham Nam','1995-02-09','237 nguyen
xi','jj7777@gmail.com','Vietnam','0837213458');
INSERT INTO `Customer` VALUES (41,11,'Nguyễn Mai Anh Nam ','2003-11-07','Thủ Đức, Tp
HCM','nguyenmaianhnam@gmail.com','Vietnam','0901798826');
INSERT INTO `Customer` VALUES (42,12,'Khang Tran','2003-07-18','11 St
French','khang1@gmail.com','French Guiana','0914589789');
INSERT INTO `Customer` VALUES (43,12,'Khang Tran','2003-07-18','11 St
French','khang1@gmail.com','French Guiana','0914589789');
INSERT INTO `Customer` VALUES (44,9,'Michelle Yang','1982-05-05','23
northcoast','MY1982@gmail.com','United States','0228901182');

```

Booking Table

```

INSERT INTO `Booking` VALUES (1,4,2,'2024-05-09','2024-05-17',8,'Confirm');
INSERT INTO `Booking` VALUES (2,5,10,'2024-05-08','2024-05-15',7,'Confirm');
INSERT INTO `Booking` VALUES (3,6,5,'2024-05-21','2024-05-25',4,'Confirm');
INSERT INTO `Booking` VALUES (4,7,10,'2024-05-09','2024-05-16',7,'Confirm');
INSERT INTO `Booking` VALUES (7,19,2,'2024-05-18','2024-05-20',2,'Confirm');
INSERT INTO `Booking` VALUES (8,23,5,'2024-05-09','2024-05-15',6,'Confirm');
INSERT INTO `Booking` VALUES (9,25,13,'2024-05-27','2024-05-31',4,'Confirm');
INSERT INTO `Booking` VALUES (10,26,5,'2024-05-02','2024-05-08',6,'Confirm');
INSERT INTO `Booking` VALUES (11,27,5,'2024-05-16','2024-05-23',7,'Confirm');
INSERT INTO `Booking` VALUES (12,28,10,'2024-05-09','2024-05-21',12,'Confirm');
INSERT INTO `Booking` VALUES (13,29,20,'2024-05-01','2024-05-13',12,'Confirm');
INSERT INTO `Booking` VALUES (14,30,2,'2024-05-23','2024-05-26',3,'Confirm');
INSERT INTO `Booking` VALUES (15,31,3,'2024-05-19','2024-05-25',6,'Confirm');
INSERT INTO `Booking` VALUES (16,32,5,'2024-07-14','2024-07-20',6,'Confirm');
INSERT INTO `Booking` VALUES (17,33,5,'2024-05-19','2024-05-25',6,'Confirm');
INSERT INTO `Booking` VALUES (18,34,8,'2024-08-18','2024-08-21',3,'Confirm');
INSERT INTO `Booking` VALUES (19,35,18,'2024-05-19','2024-05-21',2,'Confirm');
INSERT INTO `Booking` VALUES (20,36,11,'2024-09-11','2024-09-20',9,'Confirm');
INSERT INTO `Booking` VALUES (21,37,8,'2024-05-20','2024-05-22',2,'Pending');

```

```

INSERT INTO `Booking` VALUES (22,38,18,'2024-05-21','2024-05-25',4,'Confirm');
INSERT INTO `Booking` VALUES (23,39,18,'2024-05-18','2024-05-22',4,'Confirm');
INSERT INTO `Booking` VALUES (24,40,10,'2024-05-22','2024-05-31',9,'Confirm');
INSERT INTO `Booking` VALUES (26,42,19,'2024-05-20','2024-05-26',6,'Confirm');
INSERT INTO `Booking` VALUES (27,42,19,'2024-05-20','2024-05-26',6,'Confirm');
INSERT INTO `Booking` VALUES (28,44,20,'2024-05-20','2024-05-23',3,'Confirm');

```

Payment Table

```

INSERT INTO `Payment` VALUES (5,3,320.00,'2024-05-18','Card');
INSERT INTO `Payment` VALUES (6,4,1750.00,'2024-05-17','Cash');
INSERT INTO `Payment` VALUES (7,1,4000.00,'2024-05-18','Card');
INSERT INTO `Payment` VALUES (8,2,1750.00,'2024-05-18','Card');
INSERT INTO `Payment` VALUES (9,7,1000.00,'2024-05-18','Card');
INSERT INTO `Payment` VALUES (11,8,480.00,'2024-05-16','Card');
INSERT INTO `Payment` VALUES (12,9,1200.00,'2024-05-23','Cash');
INSERT INTO `Payment` VALUES (13,10,480.00,'2024-05-23','Cash');
INSERT INTO `Payment` VALUES (14,11,560.00,'2024-05-22','Cash');
INSERT INTO `Payment` VALUES (15,12,3000.00,'2024-05-22','Cash');
INSERT INTO `Payment` VALUES (16,13,1560.00,'2024-05-22','Cash');
INSERT INTO `Payment` VALUES (17,14,1500.00,'2024-05-22','Membership');
INSERT INTO `Payment` VALUES (18,15,2100.00,'2024-05-21','Card');
INSERT INTO `Payment` VALUES (19,16,480.00,'2024-05-29','Cash');
INSERT INTO `Payment` VALUES (20,17,480.00,'2024-05-23','Cash');
INSERT INTO `Payment` VALUES (21,18,630.00,'2024-05-25','Cash');
INSERT INTO `Payment` VALUES (22,19,700.00,'2024-05-25','Membership');
INSERT INTO `Payment` VALUES (23,20,2700.00,'2024-05-25','Card');
INSERT INTO `Payment` VALUES (24,23,1400.00,'2024-05-18','Card');
INSERT INTO `Payment` VALUES (25,24,2250.00,'2024-05-20','Cash');
INSERT INTO `Payment` VALUES (26,28,390.00,'2024-05-19','Membership');
INSERT INTO `Payment` VALUES (27,26,1200.00,'2024-05-25','Cash');
INSERT INTO `Payment` VALUES (28,27,1200.00,'2024-05-31','Cash');
INSERT INTO `Payment` VALUES (29,27,1200.00,'2024-05-21','Card');
INSERT INTO `Payment` VALUES (31,22,1400.00,'2024-05-25','Card');
INSERT INTO `Payment` VALUES (32,22,1400.00,'2024-05-25','Card');

```

User Table

```

INSERT INTO `User` VALUES (1,'Manager','nothing','!@#%$%^&*()Qwerty123');
INSERT INTO `User` VALUES (2,'nam','test@gmail.com','123');
INSERT INTO `User` VALUES (3,'vietnam','yuenan@gmail.com','abcxyz');
INSERT INTO `User` VALUES (4,'Heius','asasa@gmail.com','123456');
INSERT INTO `User` VALUES (5,'PhuongTran','nypd.vn.swat@gmail.com','1234');
INSERT INTO `User` VALUES (6,'Tommy','12345@gmail.com','12345');
INSERT INTO `User` VALUES (7,'abcxyz','abcxyz@gmail.com','abcxyz');
INSERT INTO `User` VALUES (8,'man','tset-3002@gmail.com','123');
INSERT INTO `User` VALUES (9,'Johny','nganhminh23@gmail.com','01qD0110');
INSERT INTO `User` VALUES (10,'Anhnam','nguyenmaianhnam@gmail.com','nguyenmaianhnam');
INSERT INTO `User` VALUES (11,'camcam71103','anhnam11b7@gmail.com','nguyenmaianhnam');
INSERT INTO `User` VALUES (12,'Khang12345','khang1@gmail.com','12345');

```

3.6. Database Query

3.6.1. Query Analysis

Most queries shown below are in the same form of this structure to ensure the user input is sanitized, where each question mark shows it is parameterized to suit the content user data:

- **INSERT INTO VALUES:** to insert data into the tables
- **SELECT:** to show the columns for return
- **FROM:** to specify the table use
- **INNER JOIN:** to join the table for more linked data query
- **WHERE:** to set a condition for each query
- **ORDER BY:** to sort the results in ascending order
- **GROUP BY:** to pack the same data into groups

3.6.2. Web Interface

a.Login Page:

Check login authentication.

```
SELECT * FROM User WHERE Email = '$Email' AND Password = '$Password'
```

Login successfully, create a new session value by email.

```
$_SESSION['usermail'] = $Email
```

b.Register account:

Check username or email exist.

```
SELECT * FROM User WHERE Email = '$Email' OR Username = '$Username'
```

Register the new one if username or email available.

```
INSERT INTO User (Username, Email, Password) VALUES ('$Username', '$Email', '$Password')
```

c.Reservation:

After login successfully, website get the User ID of this account.

```
SELECT UID FROM User WHERE email= '".$_SESSION['usermail']."'
```

`$_SESSION["UID"]` is assigned to `$GetUID`

```
$uid = $row["UID"];  
$_SESSION["UID"] = $uid;
```

Retrieve the data if the room type is available.

Note: (%d)= Integer value

```
SELECT Room.RoomNumber FROM Room, RoomType WHERE Room.Status='Ready' AND  
RoomType.Name='$RoomType' AND Room.TypeID=RoomType.TypeID AND  
RoomType.Capacity=(%d)
```

The type of room is available, record the new customer information for this.

```
INSERT INTO Customer (UID, CName, DateOfBirth, Address, Email, Country,  
Phone) VALUES ($getUID, '$Name', '$Birth', '$Address', '$Email',  
'$Country', '$Phone')
```

Retrieve CustomerID after inserting the new record.

```
SELECT CustomerID FROM Customer WHERE Email = '$Email' AND Phone =  
'$Phone'
```

Link CustomerID to Booking table and make new reservation.

```
INSERT INTO Booking  
(CustomerID,RoomNumber,CheckinDate,CheckoutDate,NoOfDays,Status) VALUES  
($nocid,'$RoomAvailable','$cin','$cout',datediff('$cout','$cin'))
```

d.Dashboard

Retrieve data what type of room were booked by guest.

```
SELECT Booking.BookingID, Customer.CName, Customer.Email,  
Customer.Country, Customer.Phone, Customer.Address, RoomType.Name,  
RoomType.Capacity, Booking.CheckinDate, Booking.CheckoutDate,  
Booking.NoOfDays, Booking.Status FROM Booking, Customer, Room, RoomType  
WHERE Booking.CustomerID = Customer.CustomerID AND Customer.UID = $s_uid  
AND Booking.RoomNumber = Room.RoomNumber AND Room.TypeID =  
RoomType.TypeID
```

If the reservation has not been approved(status: pending), user can cancel the reservation. It is executed the query respectively.

+Get BookingID from the chosen booking.

```
<a href='roombookdelete.php?id="'. $res['BookingID'] ."'><button class='btn btn-danger'>Cancel</button>
```

+Pass BookingID value and get CustomerID what reservation is pending.

```
SELECT CustomerID FROM Booking WHERE Booking.BookingID = $bookid AND  
Booking.Status = 'Pending'
```

+Assign CID value by CustomerID

```
$cid = (int)$row["CustomerID"];
```

+Delete record from Booking table.

```
DELETE FROM Booking WHERE Booking.BookingID = $bookid
```

+Delete record from Customer table

```
DELETE FROM Customer WHERE Customer.CustomerID = $cid
```

3.6.3. Application Interface

a.Login interface:

Check manager credential.

```
SELECT * FROM Management WHERE EmpUsername='?' AND PASSWORD='?'
```

b.Dashboard management:

Show employee name who is logged in.

```
SELECT Employee.LastName FROM HMS.Employee WHERE Employee.EmployeeID  
IN (SELECT EmployeeID FROM Management WHERE EmpUsername = '?')
```

c.Manage employee:

Retrieve employee data in the table.

```
SELECT EmployeeID, FirstName, MiddleName, LastName, Role, DateOfBirth,  
HireDate, Phone, Email FROM HMS.Employee
```

Add a new employee.

```
insert into HMS.Employee (FirstName, MiddleName, LastName, Role,  
DateOfBirth, HireDate, Phone, Email) values ('?', '?', '?', '?', '?', '?', '?', '?')
```

Remove a employee.

```
DELETE FROM HMS.Employee WHERE EmployeeID=? AND Role!='Manager'
```

Modify a employee information.

```
UPDATE HMS.Employee SET "
```

```
+ "FirstName='?',"  
+ "MiddleName='?',"  
+ "LastName='?',"  
+ "Role='?',"  
+ "DateOfBirth='?',"  
+ "HireDate='?',"  
+ "Phone='?',"  
+ "Email='?' "  
+ "WHERE EmployeeID=?"
```

d. Manage RoomType:

Retrieve data into the table.

```
SELECT RoomType.TypeID, RoomType.Name, RoomType.Description,  
RoomType.Price, RoomType.Capacity, Room.RoomNumber, Room.Status FROM  
RoomType, Room WHERE RoomType.TypeID = Room.TypeID
```

Add a new available room for hotel.

```
insert into HMS.RoomType (Name, Description, Price, Capacity) value  
('?', '?', '?', '?')
```

```
insert into Room(TypeID) value ((select max(RoomType.TypeID) from RoomType  
ORDER BY RoomType.TypeID DESC LIMIT 1))
```

Remove a room for hotel.

```
DELETE FROM HMS.Room WHERE RoomNumber=? AND Status!='Used'
```

Update room status.

```
update Room set Status = '?' where RoomNumber = ?
```

e. Booking:

Add a new reservation:

+Insert a record about a new customer.

```
insert into HMS.Customer (UID, CName, DateOfBirth, Address, Email, Country,  
Phone) values (1, '?', '?', '?', '?', '?', '?')
```

Note: **UID = 1** means manager create a new customer information due to UID cannot be blank.

+Get CustomerID after adding a current customer information.

```
Select CustomerID from Customer WHERE CName = '?' AND DateOfBirth = '?' AND  
Address = '?' AND Email = '?' AND Country = '?' AND Phone= '?'
```

+Insert a record about reservation for new customer after getting CustomerID.

```
insert into HMS.Booking (CustomerID, RoomNumber, CheckinDate,
CheckoutDate, NoOfDays, Status) values (?, ?, '?', '?', '?', '? ')
```

Edit a reservation:

+Get data from Booking, Customer

```
SELECT Customer.CName, "
    + "Customer.DateOfBirth, Customer.Address, "
    + "Customer.Email, "
    + "Customer.Country, Customer.Phone, Booking.RoomNumber, "
    + "Booking.NoOfDays, Booking.CheckinDate, Booking.CheckoutDate,
Booking.Status "
    + "FROM Booking, Customer, Room, RoomType "
    + "WHERE Booking.CustomerID = Customer.CustomerID "
    + "AND Booking.RoomNumber = Room.RoomNumber "
    + "AND Room.TypeID = RoomType.TypeID "
    + "AND Booking.BookingID =" + model.getValueAt(getRow(), 0))
```

Get BookingID for

```
select BookingID from HMS.Booking where CustomerID = ?
```

Remove booking

```
delete from HMS.Booking where Booking.BookingID = ?
```

f. Payment:

Choosing:

```
select RoomType.Price, RoomType.Capacity, RoomType.Description,
Booking.NoOfDays from RoomType, Room, Booking, Payment"
    + " where Booking.BookingID =" + model.getValueAt(row, 4) + "
and Room.RoomNumber = Booking.RoomNumber "
    + "and Room.TypeID = RoomType.TypeID
```

Update:

```
update Payment set Amount = "+AmountBox.getText()+", PaymentDate =
"+formatter.format(PayDateBox.getDate())+", "
        + "PaymentMethod =
"+ChoosePayMethod.getSelectedItem()+" "
        + "where Payment.PaymentID = "+model.getValueAt(getRow(),
3)
```

4. Connection between SQL and Java

Due to the need for synchronization, the team has set up an Ubuntu server running MySQL on Azure to facilitate collaboration and data import.

```
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
 * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this template
 */
package db;
import java.sql.*;
/**
 *
 * @author WOW
 */
public class connectdb {
    public static Connection getCon(){
        try{
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection("jdbc:mysql://20.2.128.32:3306/HMS?useSSL=false","pdm","hcmiu123");
            return con;
        }
        catch (Exception e){
            return null;
        }
    }
}
```

Image 4: Connection Database

This is how to connect to a SQL database using Java, using the MySQL Java module.

SQL server connection link to app and web. The group server's IP address is 20.2.128.32, port is 3306, **HMS** is the database name, **pdm** is the user credential, **hcmiu123** is the server's MySQL password.

Web interface part ⇒ the team uses PHP to connect to the database.

Project link: [Github](#)

```
<?php

$server = "20.2.232.152";
$username = "pdm";
$password = "hcmiu123";
$database = "HMS";

$conn = mysqli_connect($server,$username,$password,$database);

if(!$conn){
    die("<script>alert('connection Failed.')</script>");
} ?>
```

Image 5: Connection Database

PHP has the MySQL library ⇐ easily connects to the server's database.

Source: [hotel_mgmt/htdocs/config.php at main · qfc88/hotel_mgmt \(github.com\)](https://github.com/qfc88/hotel_mgmt/blob/main/htdocs/config.php)

5. Demo App Java

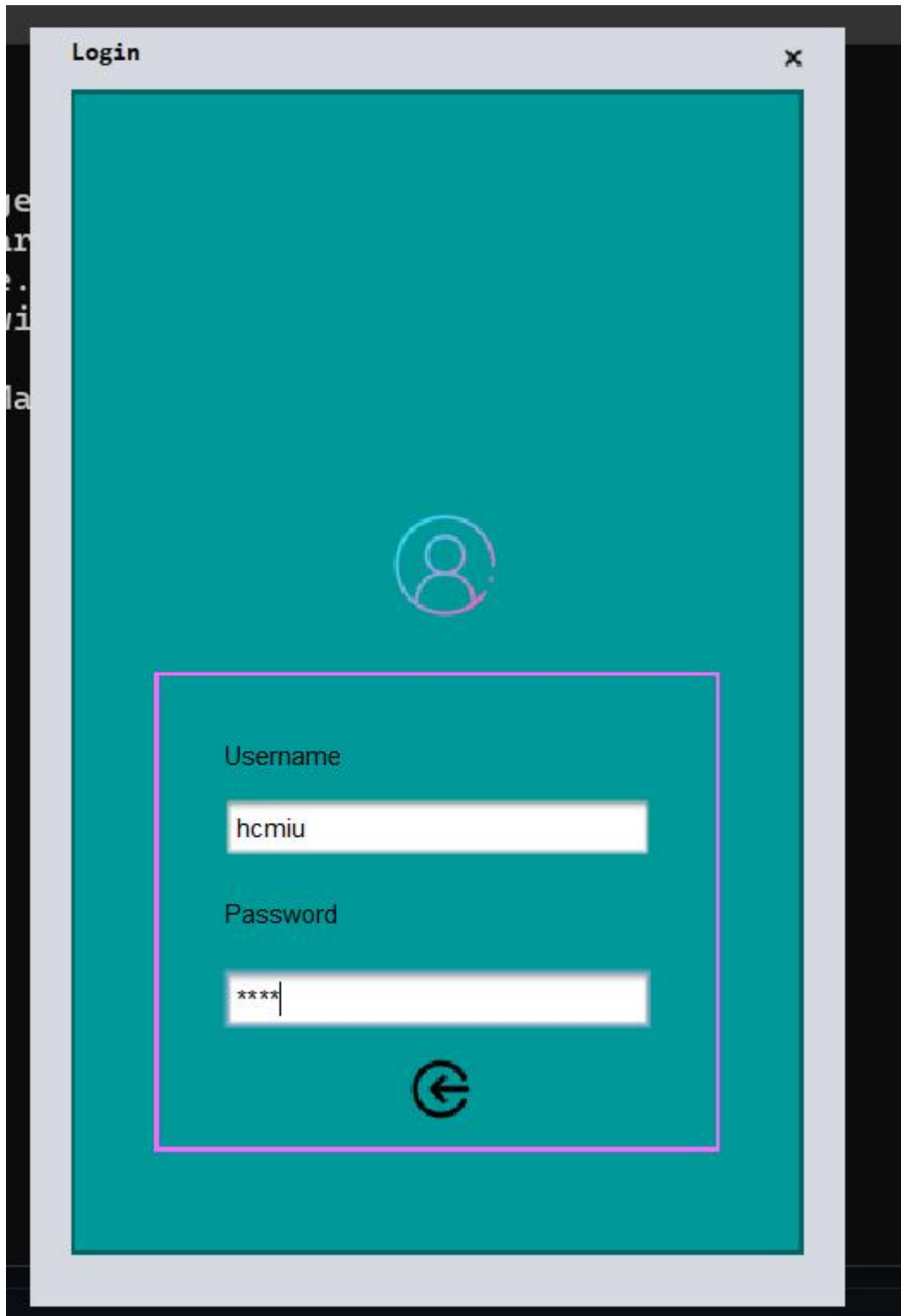


Image 5: (Login interface)

user: hcmiu, pass: 1234

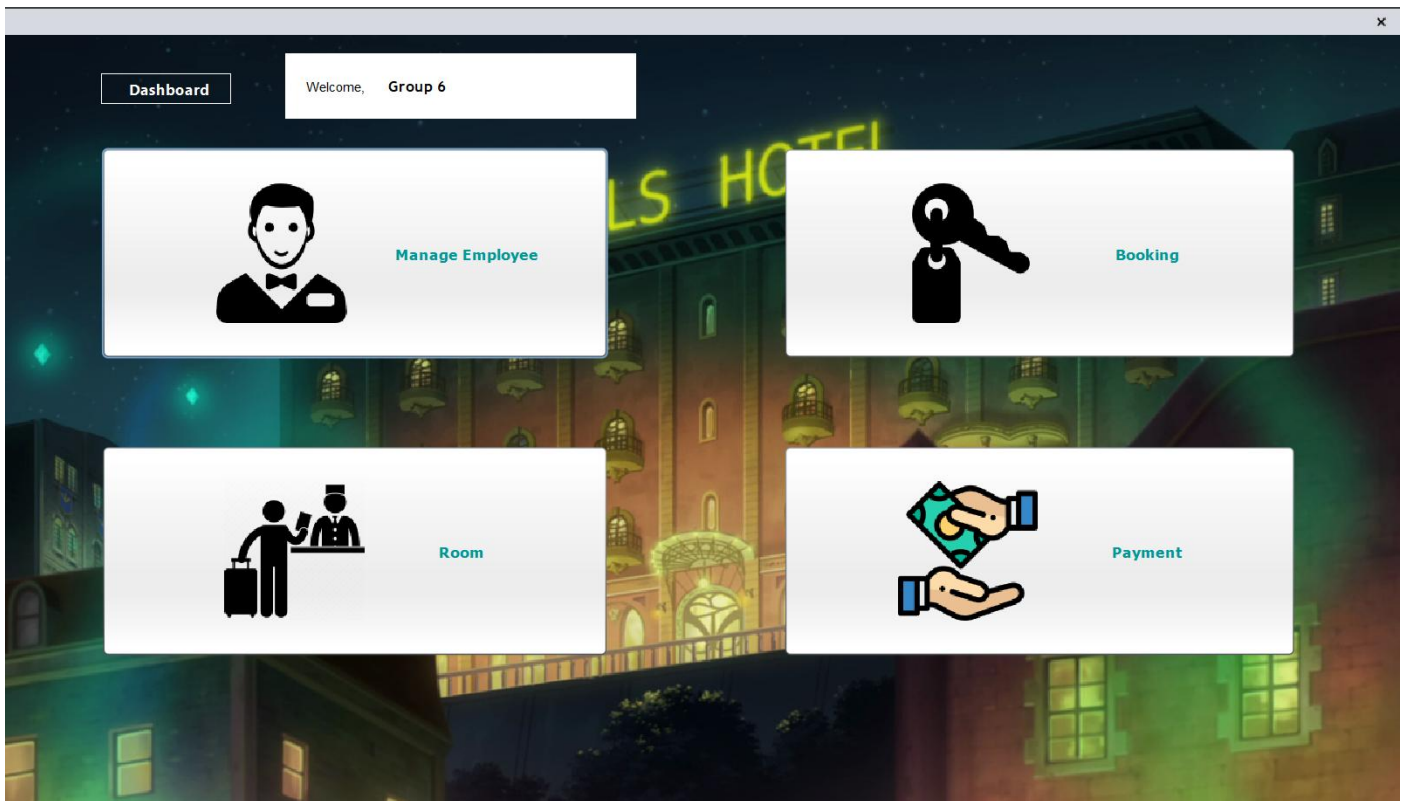
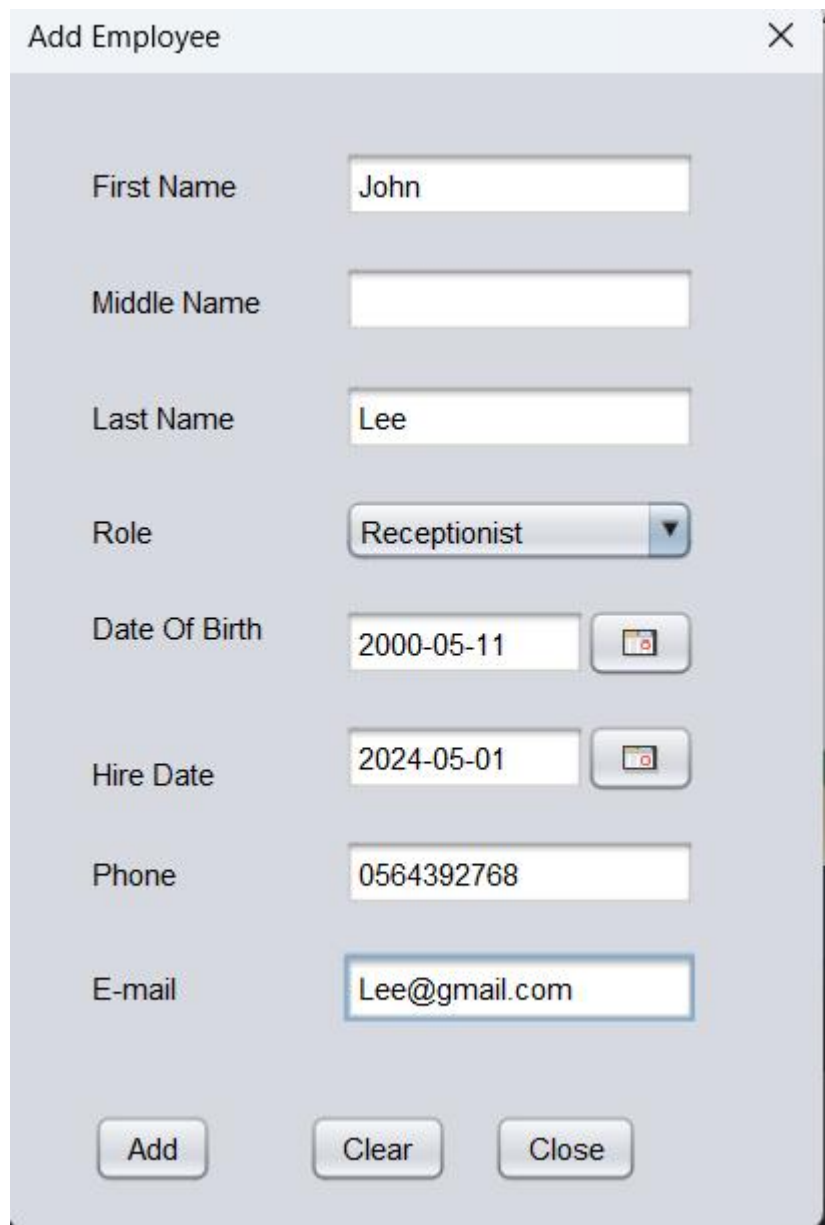


Image 6: (Dashboard Panel)



The image shows a web application window titled "Add Employee" with a close button (X) in the top right corner. The form contains the following fields and controls:

- First Name:** Text input field containing "John".
- Middle Name:** Empty text input field.
- Last Name:** Text input field containing "Lee".
- Role:** Dropdown menu with "Receptionist" selected.
- Date Of Birth:** Text input field containing "2000-05-11" and a calendar icon button.
- Hire Date:** Text input field containing "2024-05-01" and a calendar icon button.
- Phone:** Text input field containing "0564392768".
- E-mail:** Text input field containing "Lee@gmail.com".

At the bottom of the form are three buttons: "Add", "Clear", and "Close".

Image 7: (Add employee form)

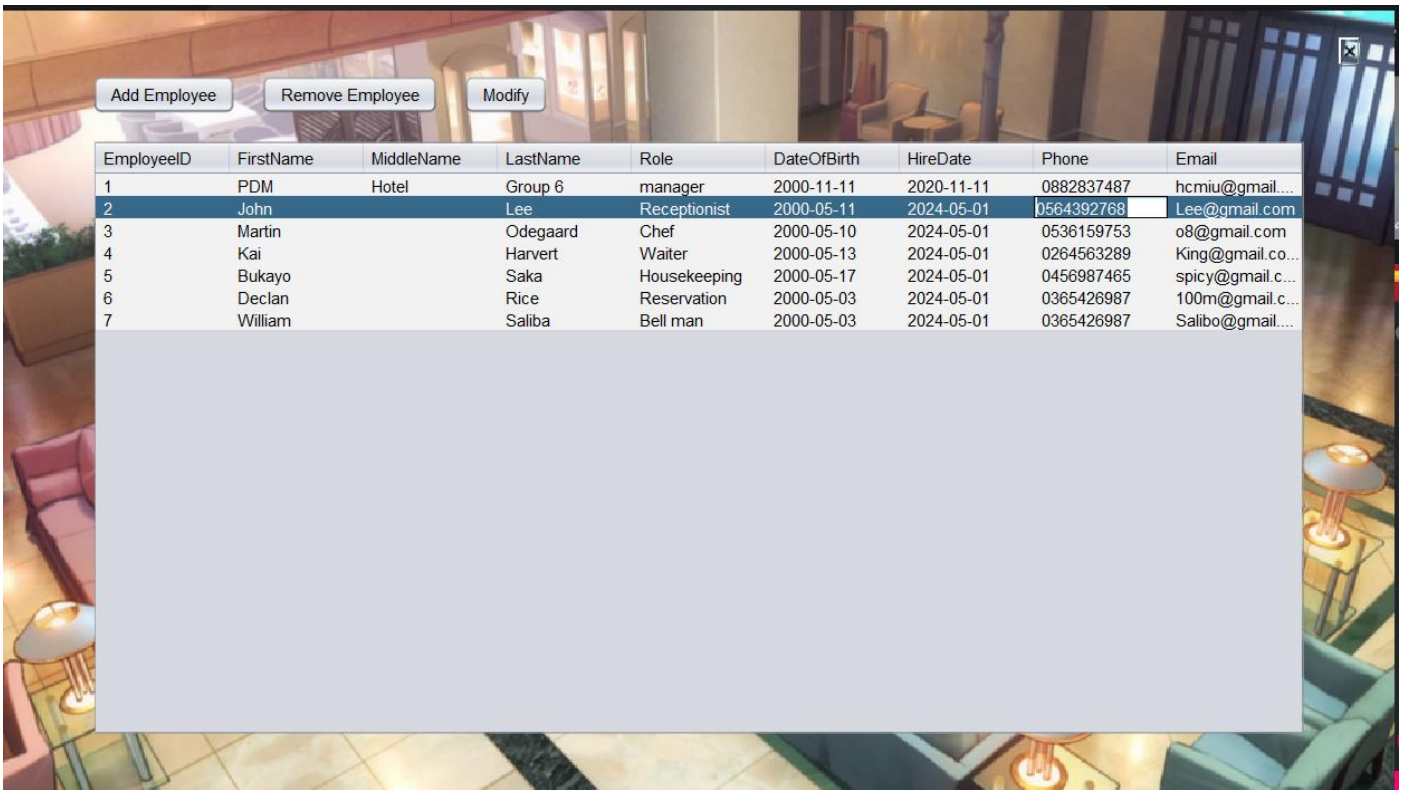


Image 8: (Detail of Employee)

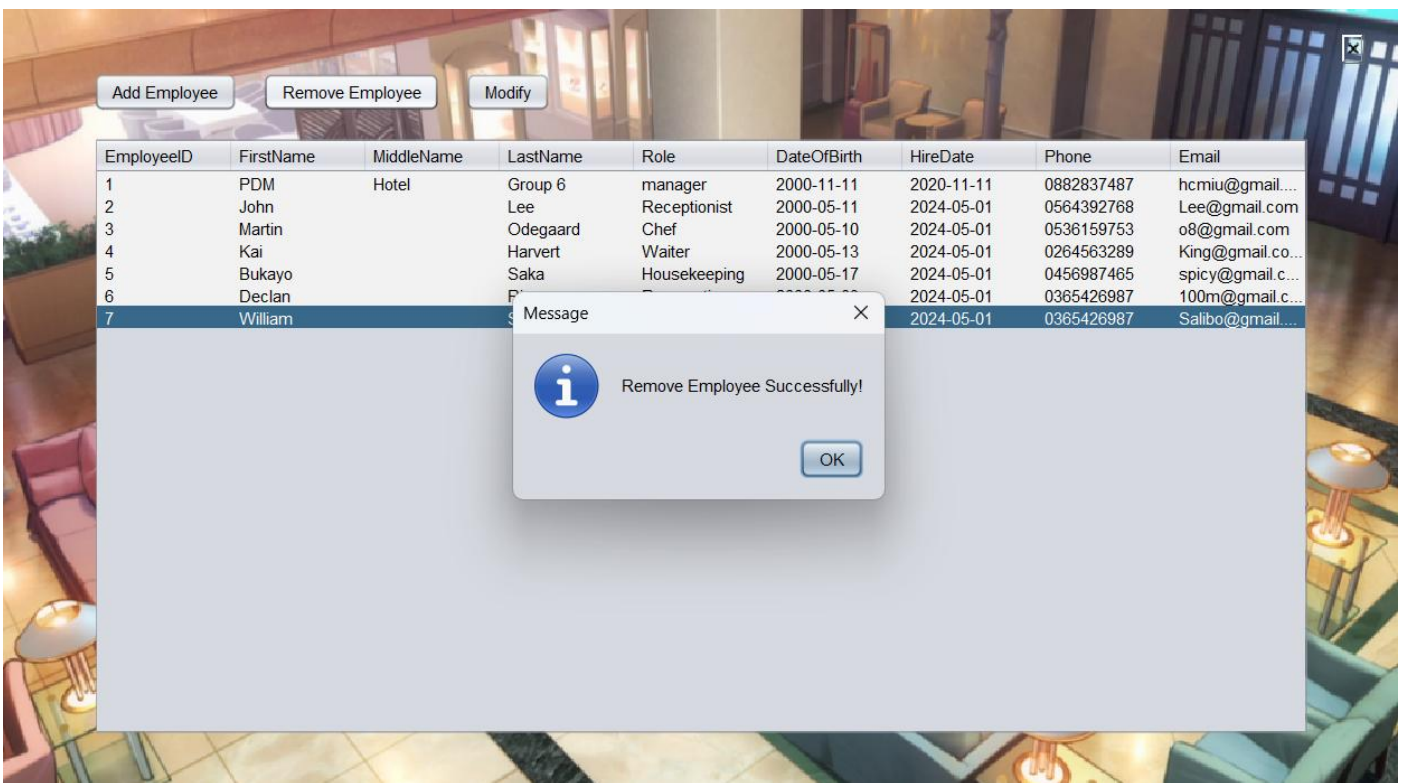


Image 9: (Remove employee form)

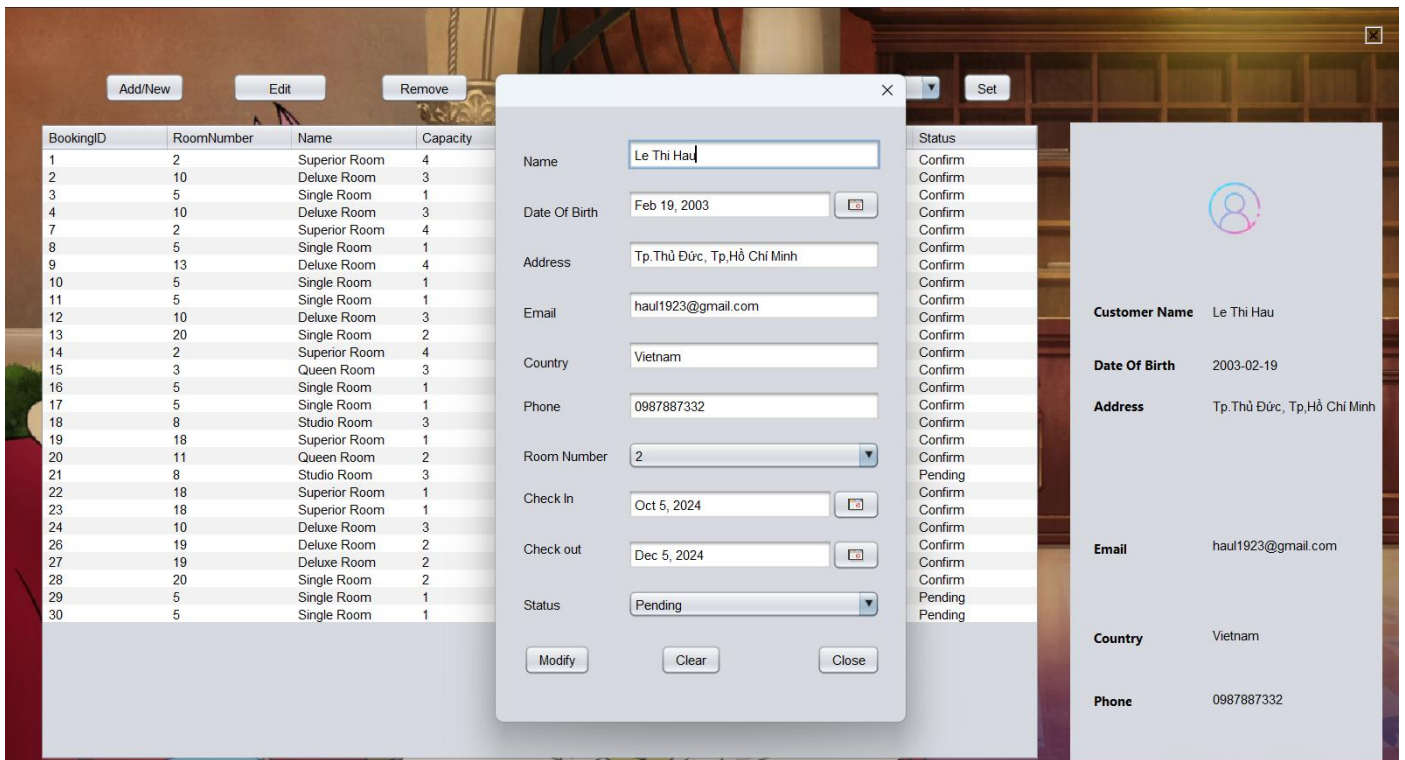


Image 12:(Customer's information management)

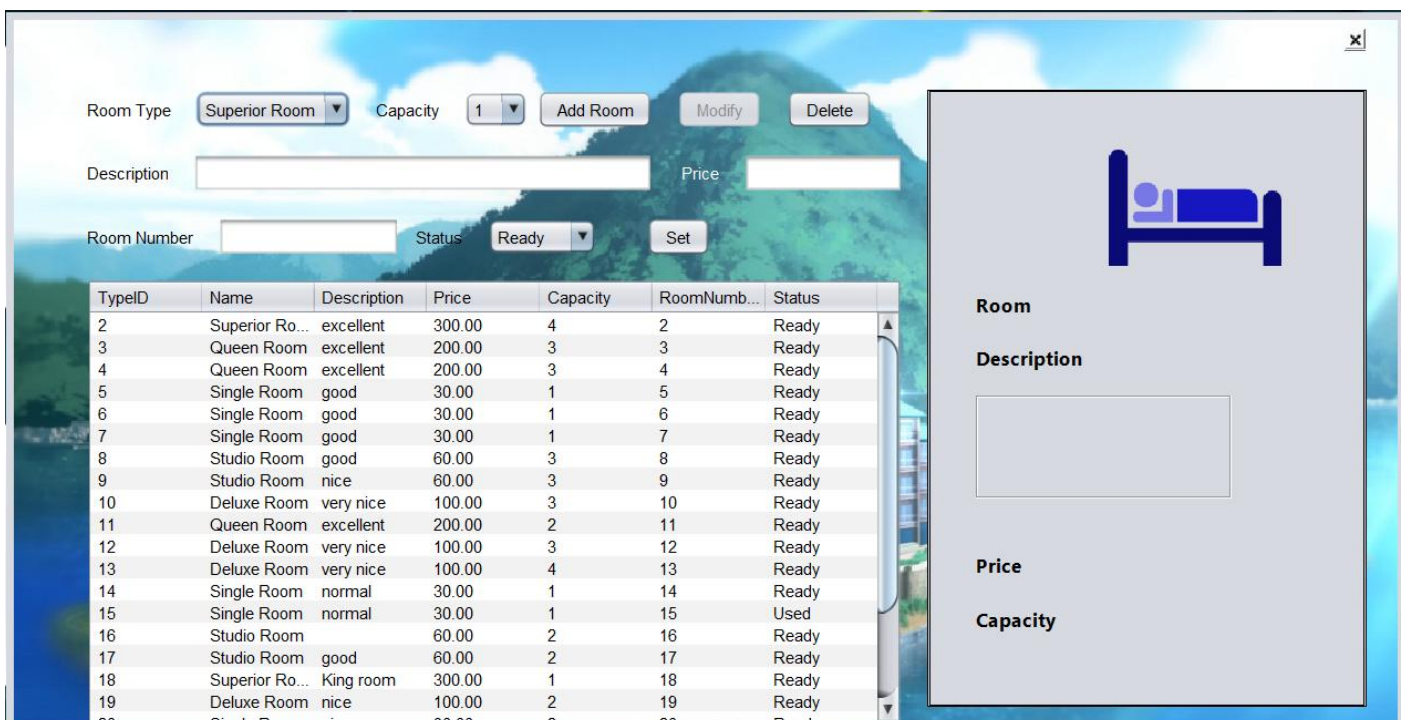


Image 13: (Details of Hotel Room)

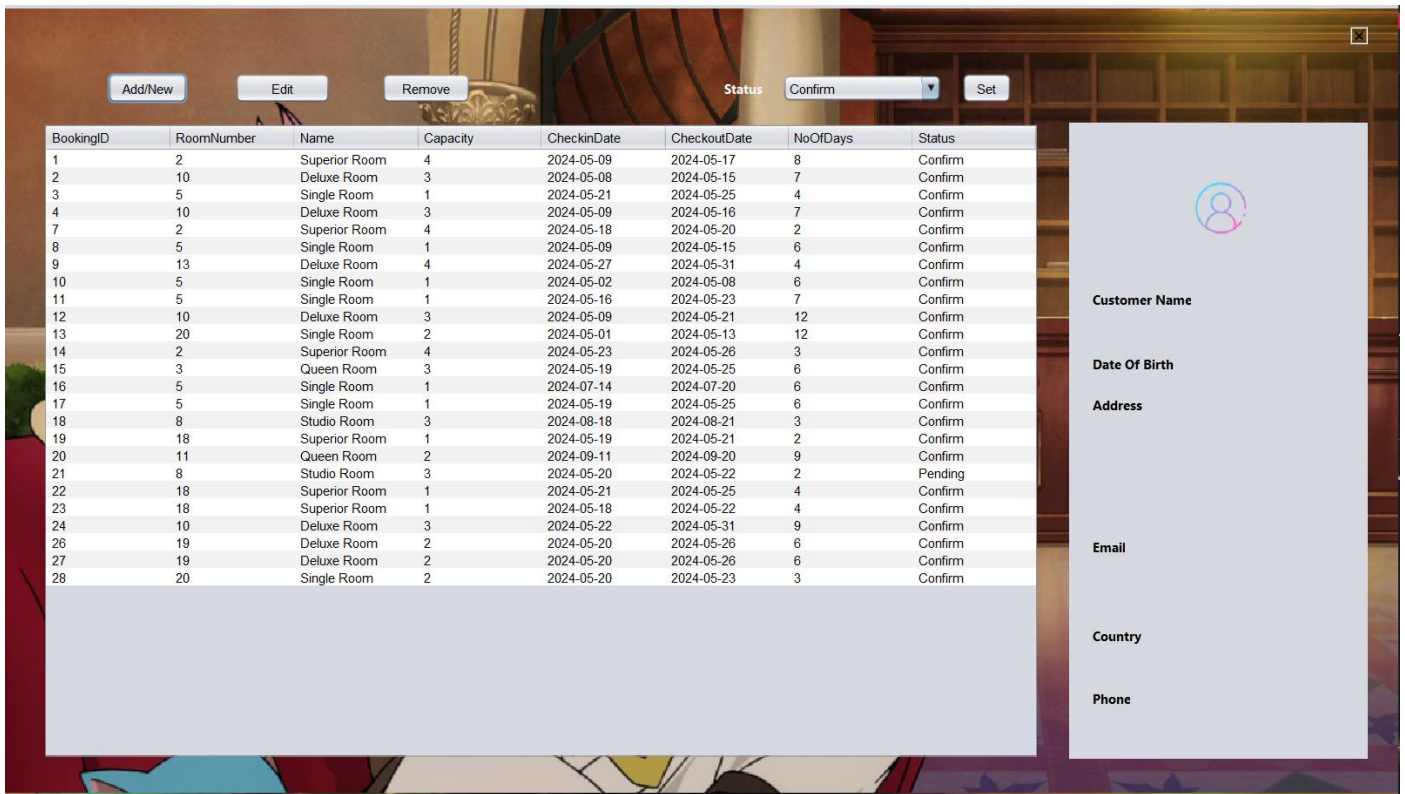


Image 14:(List Reservation)

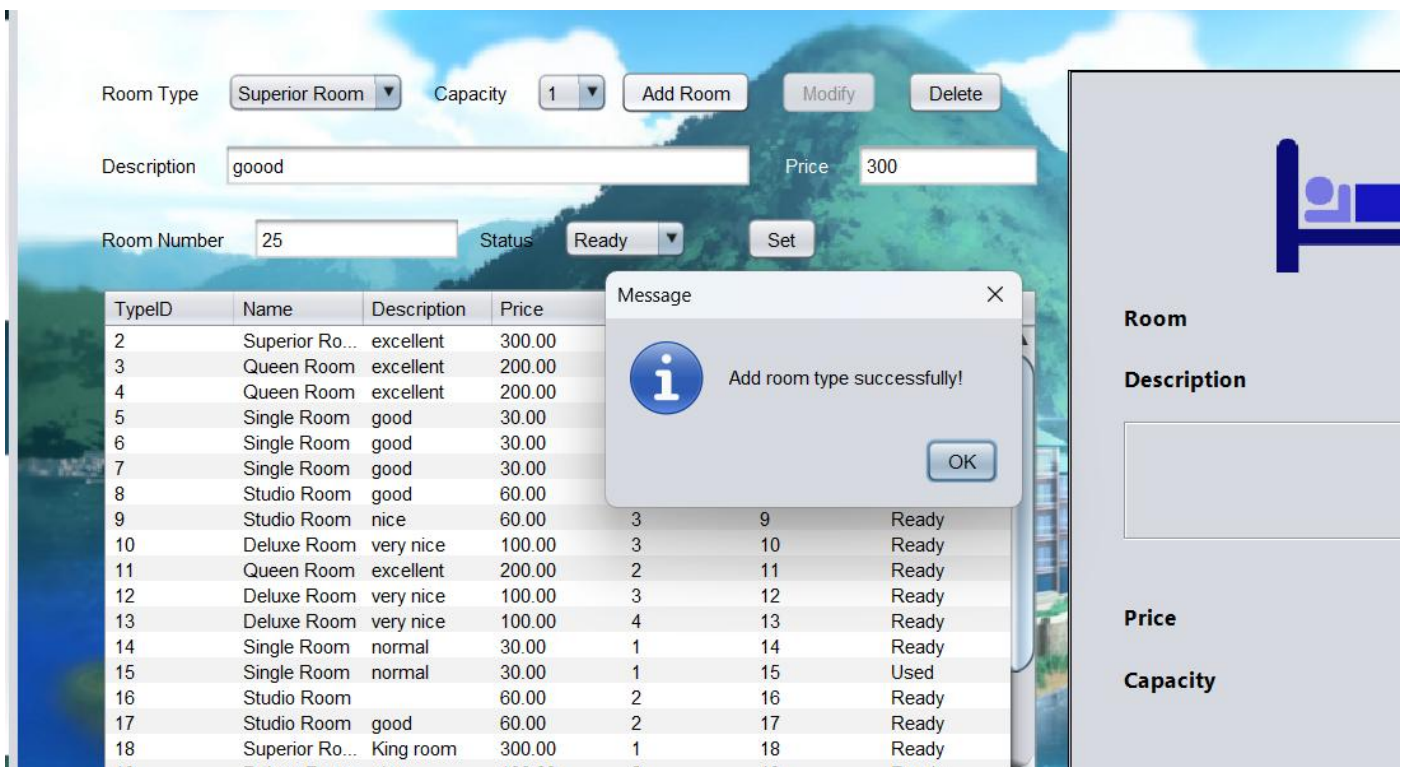


Image 15:(RoomType)

CName	Name	Capacity	PaymentID	BookingID	Amount	PaymentDate	PaymentMethod
Gabriel Jesus	Superior Room	4	7	1	4000.00	2024-05-18	Card
Thomas Partey	Deluxe Room	3	8	2	1750.00	2024-05-18	Card
Nguyen Nguyen	Single Room	1	5	3	320.00	2024-05-18	Card
Gabriel Martinelli	Deluxe Room	3	6	4	1750.00	2024-05-17	Cash
Viet Nam	Superior Room	4	9	7	1000.00	2024-05-18	Card
Zichenko	Single Room	1	11	8	480.00	2024-05-16	Card
William	Deluxe Room	4	12	9	1200.00	2024-05-23	Cash
Gabriel	Single Room	1	13	10	480.00	2024-05-23	Cash
Jorginho	Single Room	1	14	11	560.00	2024-05-22	Cash
Benjamin White	Deluxe Room	3	15	12	3000.00	2024-05-22	Cash
David Raya	Single Room	2	16	13	1560.00	2024-05-22	Cash
Alice	Superior Room	4	17	14	1500.00	2024-05-22	Membership
Eric Krylova	Queen Room	3	18	15	2100.00	2024-05-21	Card
Đào Kim Tâm	Single Room	1	19	16	480.00	2024-05-29	Cash
Rahmana Viderman	Single Room	1	20	17	480.00	2024-05-23	Cash
Harald Andreuzzi	Studio Room	3	21	18	630.00	2024-05-25	Cash
Julia Kaitainen	Superior Room	1	22	19	700.00	2024-05-25	Membership
Mee Pan	Queen Room	2	23	20	2700.00	2024-05-25	Card
Bowman Brownlock	Superior Room	1	31	22	1400.00	2024-05-25	Card
Bowman Brownlock	Superior Room	1	32	22	1400.00	2024-05-25	Card
Johnny Deep	Superior Room	1	24	23	1400.00	2024-05-18	Card
Pham Nam	Deluxe Room	3	25	24	2250.00	2024-05-20	Cash
Khang Tran	Deluxe Room	2	27	26	1200.00	2024-05-25	Cash
Khang Tran	Deluxe Room	2	28	27	1200.00	2024-05-31	Cash
Khang Tran	Deluxe Room	2	29	27	1200.00	2024-05-21	Card
Michelle Yang	Single Room	2	26	28	390.00	2024-05-19	Membership

Revenue 35130 \$

Customer Name

Room Type

Capacity

Description

Payment Date

Payment Method

Amount

Set

Image 16:(Payment Gateway)

CName	Name	Capacity	PaymentID	BookingID	Amount	PaymentDate	PaymentMethod
Gabriel Jesus	Superior Room	4	7	1	4000.00	2024-05-18	Card
Thomas Partey	Deluxe Room	3	8	2	1750.00	2024-05-18	Card
Nguyen Nguyen	Single Room	1	5	3	320.00	2024-05-18	Card
Gabriel Martinelli	Deluxe Room	3	6	4	1750.00	2024-05-17	Cash
Viet Nam	Superior Room	4	9	7	1000.00	2024-05-18	Card
Zichenko	Single Room	1	11	8	480.00	2024-05-16	Card
William	Deluxe Room	4	12	9	1200.00	2024-05-23	Cash
Gabriel	Single Room	1	13	10	480.00	2024-05-23	Cash
Jorginho	Single Room	1	14	11	560.00	2024-05-22	Cash
Benjamin White	Deluxe Room	3	15	12	3000.00	2024-05-22	Cash
David Raya	Single Room	2	16	13	1560.00	2024-05-22	Cash
Alice	Superior Room	4	17	14	1500.00	2024-05-22	Membership
Eric Krylova	Queen Room	3	18	15	2100.00	2024-05-21	Card
Đào Kim Tâm	Single Room	1	19	16	480.00	2024-05-29	Cash
Rahmana Viderman	Single Room	1	20	17	480.00	2024-05-23	Cash
Harald Andreuzzi	Studio Room	3	21	18	630.00	2024-05-25	Cash
Julia Kaitainen	Superior Room	1	22	19	700.00	2024-05-25	Membership
Mee Pan	Queen Room	2	23	20	2700.00	2024-05-25	Card
Bowman Brownlock	Superior Room	1	31	22	1400.00	2024-05-25	Card
Bowman Brownlock	Superior Room	1	32	22	1400.00	2024-05-25	Card
Johnny Deep	Superior Room	1	24	23	1400.00	2024-05-18	Card
Pham Nam	Deluxe Room	3	25	24	2250.00	2024-05-20	Cash
Khang Tran	Deluxe Room	2	27	26	1200.00	2024-05-25	Cash
Khang Tran	Deluxe Room	2	28	27	1200.00	2024-05-31	Cash
Khang Tran	Deluxe Room	2	29	27	1200.00	2024-05-21	Card
Michelle Yang	Single Room	2	26	28	390.00	2024-05-19	Membership

Revenue 35130 \$

Customer Name **Pham Nam**

Room Type **Deluxe Room**

Capacity 3

Description very nice

Payment Date

Payment Method

Amount

Set

Image 17:(Payment Gateway)

6. Demo Web

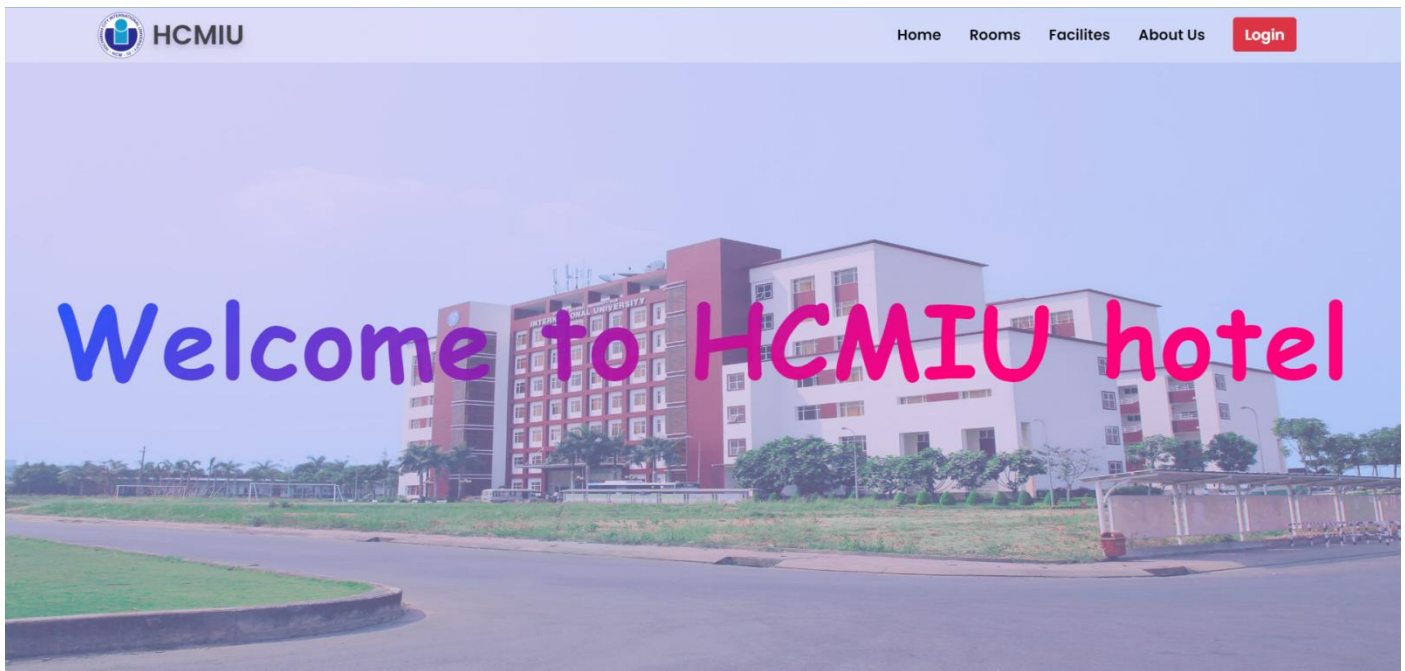


Image 18: (Welcome Interface)

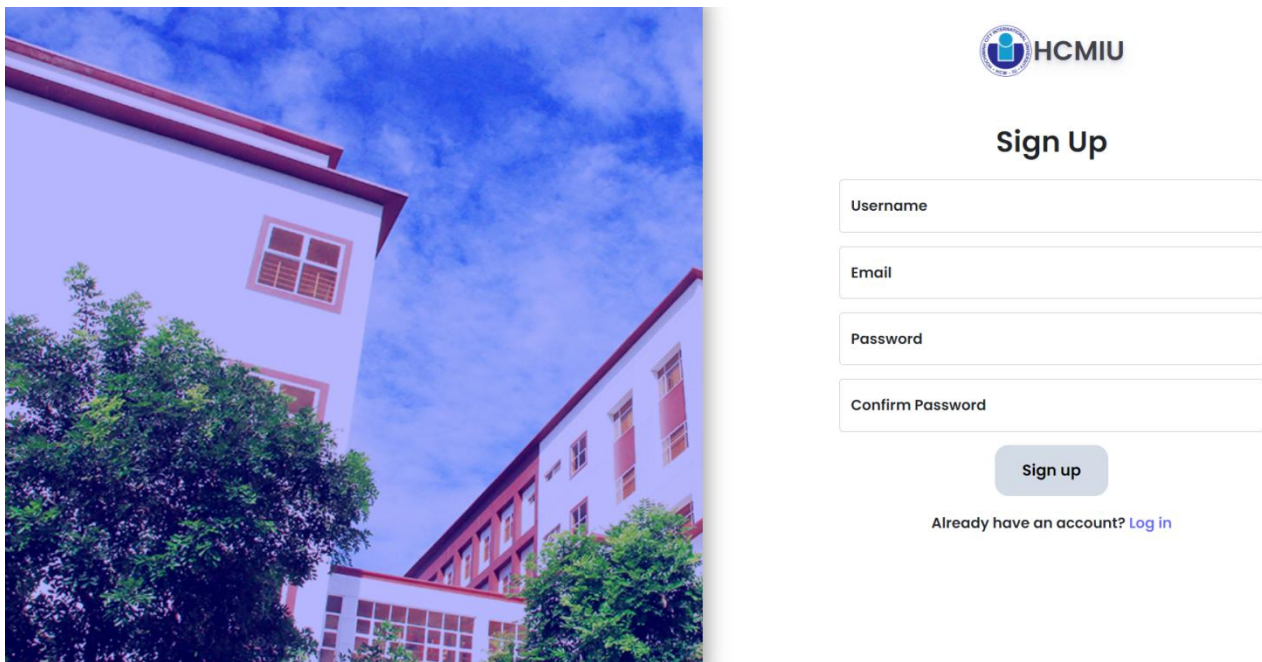


Image 19:(Registration Page)



Log In

Log in

Don't have an account? [sign up](#)

Image 20: (Login Form)

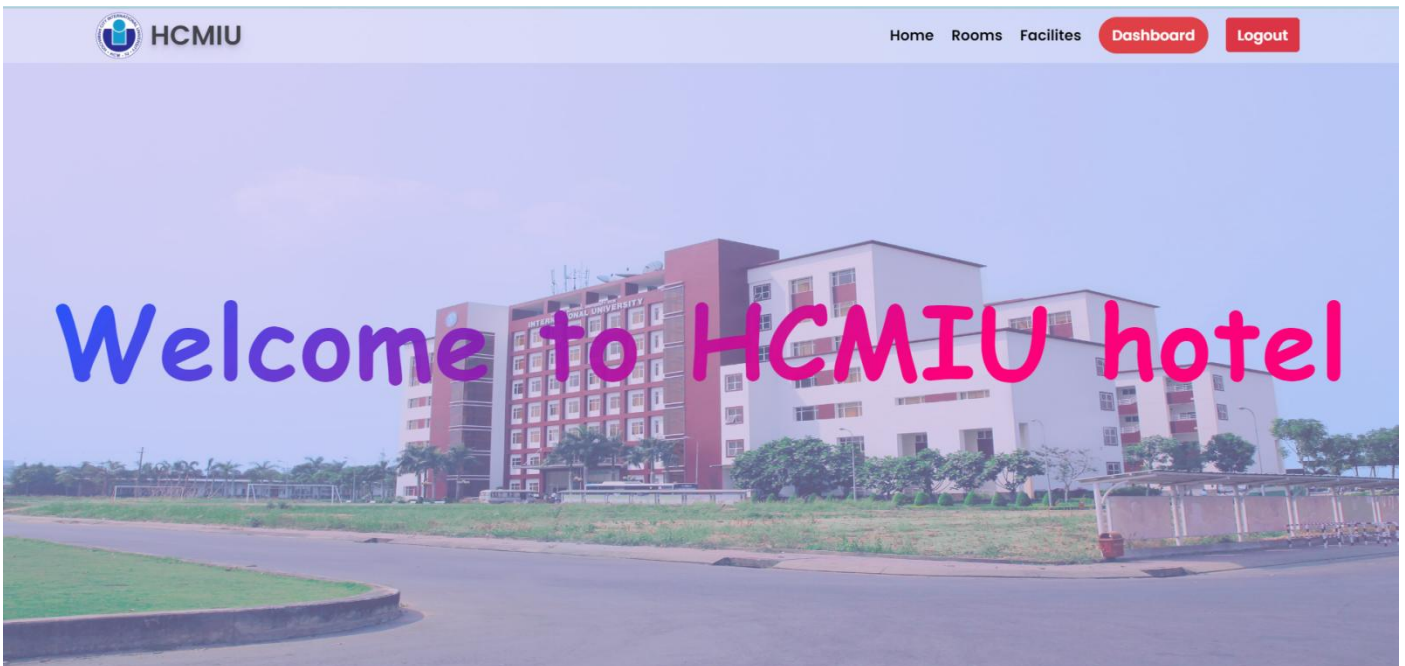


Image 21:(Successful Login)

The screenshot shows a 'RESERVATION' form with two main sections: 'Guest information' and 'Reservation information'. The 'Guest information' section includes fields for Name (Nguyễn Mai Anh Nam), Email (anhnam11b7@gmail.com), Country (Vietnam), Birth date (07/11/2003), Phone number (0901798826), and Address (Thủ Đức, Tp HCM). The 'Reservation information' section includes Room type (SUPERIOR ROOM), Number of guests (2), Check-In date (14/02/2024), and Check-Out date (16/02/2024). A 'Submit' button is located at the bottom center of the form. The background shows a hotel room and a 'Superior' room card.

Image 22: (Hotel Booking Form)

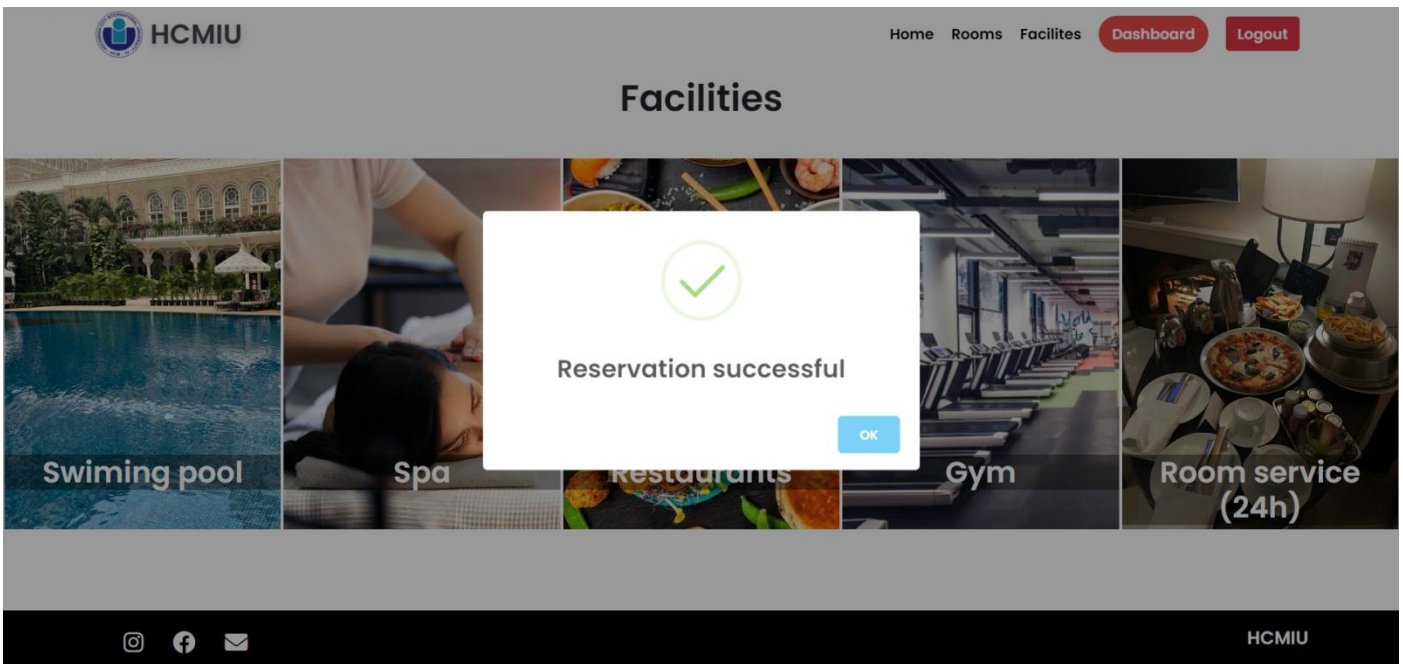


Image 23: (Successful Reservation)

 HCMIU Logout												
Dashboard search...												
ID	Name	Email	Country	Phone	Address	Type of Room	Type of Bed	Check-In	Check-Out	No of Day	Status	Action
25	Nguyễn Mai Anh Nam	nguyenmaianhnam@gmail.com	Vietnam	0901798826	Thủ Đức, Tp HCM	Deluxe Room	2	2024-02-14	2024-02-16	2	Pending	Cancel

Image 24: (Dashboard - Booking Page)

For security reasons (there is a server IP address in here), teachers and teaching assistants want to check so the group sets up public GitHub or add teachers or TA’ accounts into GitHub repository. (https://github.com/qfc88/hotel_mgmt/)

7. Conclusion and recommendation

7.1. Achieved goals

Our team is proud to announce the successful development of our hotel management system, achieving all its initial goals. Here's a breakdown of our key accomplishments:

Intuitive User Interface: We prioritized a user-friendly interface for both staff and guests. The system is designed for efficient navigation and a smooth user experience.

Our hotel management system has achieved its initial goals, delivering a comprehensive and efficient solution for hotel operations and guest experience. We are committed to ongoing development and innovation to stay ahead of the curve in the hospitality industry.

7.2. Future Work

While our hotel management system has successfully launched and met its initial goals, we're dedicated to continuous improvement. Here are some key areas where we see exciting opportunities for future development:

Strengthening Security: Data security remains a top priority. We'll be implementing two-factor authentication for an extra layer of protection for guest information and financial transactions.

Embracing Innovation: The hospitality industry is constantly evolving. Our development team will closely monitor emerging technologies, such as:

- **Chatbots:** Chatbots can enhance communication by offering 24/7 guest support or answering frequently asked questions.

Industry Trend Awareness: We'll stay informed about industry trends to integrate functionalities that maximize guest satisfaction and operational efficiency. This could involve features like:

- **Mobile Check-In/Out:** Streamline the arrival and departure process for guests.
- **Self-Service Kiosks:** Allow guests to manage their reservations, request services, or access information independently.

By prioritizing these advancements, we are confident that our hotel management system will remain a valuable asset for years to come, ensuring a competitive edge in the ever-changing hospitality landscape.

7.3. Concluding thought

In conclusion, we are confident that our hotel management system holds significant promise for the future. We plan to continually refine the system to ensure even greater efficiency and user-friendliness for both staff and guests. This project has provided invaluable experience for our team in collaboration, communication, and problem-solving, skills that will prove beneficial in future endeavors.

We extend our sincere gratitude to all who supported the development of this system. Their guidance and expertise were instrumental in our success. We are committed to playing a positive role in the hospitality industry by streamlining operations and enhancing the guest experience. With a focus on continuous development and innovation, we are enthusiastic about the potential of our hotel management system to revolutionize the way hotels operate.

8. References:

Tushar Pankhaniya: <https://github.com/tushar-2223/>

BTech Days: <https://www.youtube.com/@BTechDays>

Stackoverflow: <https://stackoverflow.com/>

Module: <https://dev.mysql.com/downloads/connector/j/?os=26>

Module: <https://toedter.com/jcalendar/>

Module: <https://sourceforge.net/projects/finalangelsanddemons/>